Proximal Methods in Numerical Optimization

Lecture III – Proximal Gradient Methods

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⚠ these slides are under development: please email me for corrections and suggestions ⚠

Outline

Settings

Proximal gradient methods

Convergence theory

Practical aspects

Toward (quasi-)Newton methods

Structured Optimization Problem

Problem (P)

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \;\; \varphi(x) \;:=\; f(x) + g(x)$$

Smooth part f

- $f : \mathbb{R}^n \to \mathbb{R}$ differentiable
- ∇f globally/locally Lipschitz/continuous
- possibly nonconvex

Nonsmooth part g

- $g : \mathbb{R}^n \to \overline{\mathbb{R}}$ proper, lsc (unif. prox-bounded)
- prox-friendly: "simple", often convex
- $\operatorname{dom} g$ encodes constraints

Motivating Examples

Example 1: LASSO

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} \|Ax - b\|^2 + \lambda \|x\|_1$$

- f is $\mathcal{C}^\infty$, convex, $\nabla f(x) = A^\top (Ax - b)$
- g promotes sparsity
- soft-thresholding $\text{prox}_{\alpha g}$

Example 2: Constrained QP

$$\underset{x \in C}{\text{minimize}} \quad \frac{1}{2} x^\top Q x + q^\top x$$

- f is $\mathcal{C}^\infty$, quadratic
- $g = \delta_C$ (indicator of closed set C)
- $\text{prox}_{\alpha g} = \text{proj}_C$
- recovers projected gradient method

Further examples: group LASSO, nuclear norm minimization, sparse inverse covariance, neural-network training with ℓ_1 regularization, phase retrieval, matrix factorization...

Gradient Descent: A Quick Review

Problem: $\text{minimize}_x\, f(x)$ with $f \in \mathcal{C}^1$.

Gradient Descent (GD)

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

Three views of the same step:

1. **Steepest descent:** follow direction $-\nabla f(x_k)$, scaled by α_k
2. **Linearization:** $x_{k+1} \in \arg\min_x \left[f(x_k) + \langle \nabla f(x_k), x - x_k \rangle \right]$ ... (unbounded, TR)
3. **Quadratic model:**

$$Q(x; x_k, \alpha) := f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2\alpha} \|x - x_k\|^2$$

$$\alpha_k > 0 \quad \Longrightarrow \quad x_{k+1} \in \arg\min_x Q(x; x_k, \alpha_k) = x_k - \alpha_k \nabla f(x_k)$$

The quadratic model is the key to generalizing to structured problems $f + g$.

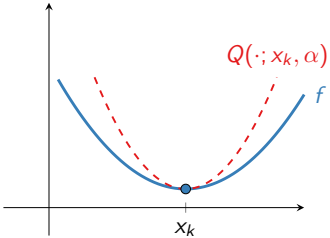
Proximal gradient methods

Quadratic Upper Models: Toward Majorization-Minimization

Suppose ∇f is (locally) L-Lipschitz, namely

$$f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2 = Q(y; x, 1/L)$$

The quadratic model $Q(\cdot; x, \alpha)$ is a (local) upper bound on f for all $\alpha \in (0, 1/L)$.



Minimizing the majorization model $Q(\cdot; x_k, 1/L) + g$:

$$\begin{aligned} x_{k+1} &\in \arg\min_x \{Q(x; x_k, 1/L) + g(x)\} \\ &= \arg\min_x \left\{ f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{L}{2} \|x - x_k\|^2 + g(x) \right\} \\ &= \arg\min_x \left\{ g(x) + \frac{L}{2} \left\| x - x_k + \frac{\nabla f(x_k)}{L} \right\|^2 \right\} = \mathrm{prox}_{\alpha g}\left(x_k - \alpha \nabla f(x_k)\right) \end{aligned}$$

where $\alpha \equiv 1/L$. This is the **proximal gradient step** for $f + g$ at x_k with stepsize α.

Proximal Gradient Method (PGM)

Algorithm: basic PGM

Choose $x^0 \in \operatorname{dom} g$, stepsize $\alpha > 0$
for $k = 0, 1, 2, \ldots$ **do**
    $x_{k+1} \in \operatorname{prox}_{\alpha g}\big(x_k - \alpha \nabla f(x_k)\big)$
end for

Interpretation:

- Half-step: *gradient* on f
- Half-step: *proximal* on g
- Operator splitting: forward-backward, explicit-implicit

Fixed-point iteration.
Characterization for **convex** g:

$$\begin{aligned} x^\star \text{ stationary} &\iff 0 \in \partial g(x^\star) + \nabla f(x^\star) \\ &\iff x^\star = \operatorname{prox}_{\alpha g}(x^\star - \alpha \nabla f(x^\star)) \end{aligned}$$

Projected Gradient as a Special Case

Let $g = \delta_C$ (indicator of closed set $C \subseteq \mathbb{R}^n$):

$$\delta_C(x) = \begin{cases} 0 & x \in C \\ +\infty & x \notin C \end{cases}$$

Then:

$$\operatorname{prox}_{\alpha\,\delta_C}(z) = \arg\min_{u \in C} \|u - z\|^2 = \operatorname{proj}_C(z)$$

Projected Gradient Descent (PGD)

$$x_{k+1} \in \operatorname{proj}_C\big(x_k - \alpha_k \nabla f(x_k)\big)$$

Examples: box constraints (ℓ_∞ ball), simplex projection, nuclear norm ball, PSD cone, complementarity constraints...

Block Coordinate Proximal Gradient

When $x = (x_1, \ldots, x_B)$ is partitioned into blocks:

$$\varphi(x) := f(x) + \sum_{b=1}^{B} g_b(x_b)$$

Block Coordinate Descent (BCD-PGM)

1: **for** $k = 0, 1, \ldots$ **do**
2: Pick block b_k
3: $(x_{b_k})_{k+1} = \mathrm{prox}_{\alpha g_{b_k}}\left((x_{b_k})_k - \alpha \nabla_{b_k} f(x_k)\right)$
4: **end for**

- Cheap per-iteration cost (partial gradient)
- Cyclic or random selection of blocks b_k for convergence to stationary points
- Suited for matrix factorization, dictionary learning, sparse NMF

Convergence theory

Global Lipschitz Gradient Assumption

Assumption (L)

∇f is L-Lipschitz continuous: there exists a finite L such that

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

- Descent lemma: $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2$ for all x, y.

$$f(x_+) \leq f(x) + \langle \nabla f(x), x_+ - x \rangle + \frac{L}{2} \|x_+ - x\|^2$$

- Prox-grad step: $x_+ \in \mathrm{prox}_{\alpha g}(x - \alpha \nabla f(x)) = \arg\min_x \left\{ g(z) + \frac{1}{2\alpha} \|z - x\|^2 \right\}$.

$$g(x_+) + \frac{1}{2\alpha} \|x_+ - x + \alpha \nabla f(x)\|^2 \leq g(x) + \frac{\alpha}{2} \|\nabla f(x)\|^2$$

$$\implies \quad \varphi(x_+) \leq \varphi(x) + \frac{\alpha L - 1}{2\alpha} \|x_+ - x\|^2$$

Global Lipschitz Gradient Assumption II

Assumption (L)

∇f is L-Lipschitz continuous: there exists a finite L such that

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \quad \forall x, y \in \mathbb{R}^n.$$

$$\varphi(x_+) \leq \varphi(x) + \frac{\alpha L - 1}{2\alpha} \|x_+ - x\|^2$$

With $\alpha = \sigma/L$, $\sigma \in (0, 1)$, there is **sufficient decrease**:

$$x_{k+1} \in \mathrm{prox}_{\alpha g}(x_k - \alpha \nabla f(x_k)) \quad \Longrightarrow \quad \varphi(x_{k+1}) \leq \varphi(x_k) - \frac{1 - \sigma}{2\alpha} \|x_{k+1} - x_k\|^2$$

Issue: L often unknown or overly conservative $\Rightarrow$ use backtracking to find good α.

Gradient Mapping

Gradient mapping

Generalized gradient for $f + g$:

$$G_\alpha(x) := \frac{x - \mathrm{prox}_{\alpha g}(x - \alpha \nabla f(x))}{\alpha} = \frac{x - x_+}{\alpha}$$

Why it matters:

- Reduces to $\nabla f(x)$ when $g = 0$ (unconstrained, smooth).
- **Stationarity:** $G_\alpha(x^\star) = 0 \iff 0 \in \nabla f(x^\star) + \partial g(x^\star)$ for **convex** g.
- $\|G_\alpha(x)\|$ is a natural *measure of stationarity* at x.

$$\varphi(x_+) \le \varphi(x) - \alpha \frac{1-\sigma}{2} \|G_\alpha(x)\|^2 = \varphi(x) - \frac{1-\sigma}{2\alpha} \|x_+ - x\|^2$$

PGM step: $x_{k+1} = x_k - \alpha G_\alpha(x_k)$.

- PGM as GD with *gradient mapping*.
- Convergence requires $\left\|G_\alpha(x^k)\right\| \to 0$.

Sufficient Decrease

Lemma (Sufficient Decrease).

Under Assumption (L), with $\alpha \in (0, 1/L)$ and some $\sigma \in (0, 1)$:

$$\varphi(x_{k+1}) \leq \varphi(x_k) - \frac{\sigma\alpha}{2} \|G_\alpha(x_k)\|^2 \leq \varphi(x_k)$$

If $\varphi := f + g$ is bounded from below by $\varphi_\star$, then starting with $x_0 \in \operatorname{dom} \varphi$ and telescoping

$$\infty > \varphi(x_0) - \varphi_\star \geq \sum_{k=0}^{K} \varphi(x_k) - \varphi(x_{k+1}) \geq \frac{\sigma\alpha}{2} \sum_{k=0}^{K} \|G_\alpha(x_k)\|^2$$

meaning that $\|G_\alpha(x_k)\| \to 0$.

Backtracking Line Search

When L is unknown or large, use **adaptive stepsize**, typically based on **backtracking**:

Backtracking: one PGM iteration

Input: x_k, $\bar{\alpha} > 0$, $\beta \in (0, 1)$
$\alpha_k \leftarrow \bar{\alpha}$
$x_{k+1} \leftarrow \mathrm{prox}_{\alpha_k g}(x_k - \alpha_k \nabla f(x_k))$
while $f(x_{k+1}) > Q(x_{k+1}; x_k, \alpha_k)$ **do**
    $\alpha_k \leftarrow \beta \alpha_k$
    $x_{k+1} \leftarrow \mathrm{prox}_{\alpha_k g}(x_k - \alpha_k \nabla f(x_k))$
end while

where $Q(y; x, \alpha) := f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2\alpha} \|y - x\|^2$ is a quadratic model of f around x.

- Adaptive to local curvature
- Guarantees sufficient decrease at each iteration
- Terminates in finite steps (since $\alpha \leq 1/L$ satisfies the condition)
- ... even for locally Lipschitz ∇f, or less [DMT22, KM22]

Convergence: Convex Case

Additional Assumption (C):

f and g are convex; $\varphi_\star = \min \varphi > -\infty$.

Theorem (Sublinear Rate, $\mathcal{O}(1/k)$).

Under (L) + (C), with fixed $\alpha = 1/L$:

$$\varphi(x_k) - \varphi_\star \leq \frac{L \|x_0 - x_\star\|^2}{2k}$$

Key steps in proof:

- Sufficient decrease + convexity of g $\implies$ one-step progress inequality.
- Telescope the sufficient-decrease inequality over k steps.
- Use convexity of φ to bound $\varphi(x_k)$ by $\frac{1}{k}\sum_j \varphi(x_j)$.

Linear Convergence Under Strong Convexity

Assumption (SC):

f is μ-strongly convex ($\mu > 0$).

Theorem (Linear Rate).

Under (L) + (SC), with $\alpha = 1/L$:

$$\|x_k - x_\star\|^2 \leq \left(1 - \frac{\mu}{L}\right)^k \|x_0 - x_\star\|^2$$

Condition number $\kappa = L/\mu$ governs the rate.

Derivation: The key contraction is

$$\|\mathcal{T}_\alpha(x) - \mathcal{T}_\alpha(y)\|^2 \leq \left(1 - \frac{\mu}{L}\right) \|x - y\|^2$$

where $\mathcal{T}_\alpha(x) := \mathrm{prox}_{\alpha g}(x - \alpha \nabla f(x))$.

Convergence: Nonconvex Case

Without convexity, we cannot expect $\varphi(x_k) \to \varphi_\star$. Instead:

Theorem (Stationarity in $\mathcal{O}(K)$).

Under Assumption (L), backtracking PGM produces:

$$\min_{0 \le k \le K} \|G_\alpha(x_k)\|^2 \le \frac{2}{\alpha_{\min} K}(\varphi(x_0) - \varphi_\star)$$

Hence $G_\alpha(x_k) \to 0$ along a subsequence.

Proof sketch:

- Sufficient decrease: $\varphi(x_{k+1}) \le \varphi(x_k) - \frac{\alpha_k}{2} \|G_{\alpha_k}(x_k)\|^2$.
- Telescope: $\sum_{k=0}^{K-1} \frac{\alpha_k}{2} \|G_{\alpha_k}(x_k)\|^2 \le \varphi(x_0) - \varphi_\star < \infty$.
- $\alpha_k \ge \alpha_{\min} := \beta/L > 0$.

If ∇f not globally Lipschitz, $\alpha_{\min}$ depends on geometry around accumulation points...

Kurdyka–Łojasiewicz (KL) Framework

For finer convergence results in the nonconvex case, we use:

KL Property.

Function φ satisfies KL at $x_\star$ if there exists a desingularizing function ψ such that

$$\psi'(\varphi(x) - \varphi(x_\star)) \, \|\partial\varphi(x)\| \geq 1$$

holds for all x near $x_\star$.

KL covers:

- Real-analytic, semi-algebraic functions (polynomials, ℓ_1-regularized, ...)
- Piecewise smooth functions
- Practically all functions encountered in ML/stats

Convergence with KL: whole sequence

Under KL property and sufficient decrease, PGM sequence **converges** to a critical point with rate determined by ψ (linear/sublinear depending on KL exponent).

Accelerated Gradient Methods: Nesterov's Idea

Can we do better than $\mathcal{O}(1/k)$ for convex problems?

Nesterov (1983): *Yes!* Using **momentum**.

Key idea — extrapolation: instead of applying prox-grad at x_k, apply it at a *momentum point / looking ahead*:

$$y_k = x_k + \tau_k(x_k - x_{k-1})$$

where $\{\tau_k\}$ is a carefully chosen sequence.

Lower bound (Nesterov, 1983)

No first-order method can achieve better than $\mathcal{O}(1/k^2)$ in general.

FISTA's revolution: Fast ISTA / PGM

Algorithm: FISTA — Beck & Teboulle [BT09]

$x_0 = y_1 \in \operatorname{dom} g,\ t_1 = 1,\ \alpha = 1/L$
for $k = 1, 2, \ldots$ **do**
  $x_k = \operatorname{prox}_{\alpha g}\big(y_k - \alpha \nabla f(y_k)\big)$
  $t_{k+1} = \frac{1}{2}\big(1 + \sqrt{1 + 4t_k^2}\big)$
  $y_{k+1} = x_k + \frac{t_k - 1}{t_{k+1}}(x_k - x_{k-1})$
end for

Theorem.

Under (L)+(C), FISTA achieves $\mathcal{O}(1/k^2)$:

$$\varphi(x_k) - \varphi_\star \leq \frac{2L\,\|x_0 - x_\star\|^2}{(k+1)^2}.$$

Momentum factor $\tau_k \equiv \frac{t_k - 1}{t_{k+1}} \to 1^-$ asymptotically, $y_k \approx 2x_k - x_{k-1}$.

- Momentum can hurt near stationary points $\rightsquigarrow$ restarting (empirically much faster, especially for ill-conditioned problems)
- Monotone variants to ensure $\varphi(x^k)$ decreasing (slightly slower in practice)

Acceleration in the Nonconvex Setting

Warning

Nesterov momentum does *not* guarantee acceleration for nonconvex φ.

iPiano [OCBP14]: inertial PGM for nonconvex f and convex g:

$$x_{k+1} = \mathrm{prox}_{\alpha_k g}\big(x_k - \alpha_k \nabla f(x_k) + \beta_k (x_k - x_{k-1})\big)$$

Convergence under KL property + sufficient decrease condition on (α_k, β_k).

APGM-NC [LL15]: Nesterov acceleration with monitoring, for nonconvex f and nonconvex g. Achieves global convergence to stationary points in general, and complexity $\mathcal{O}(1/k^2)$ for convex problems.

Practical aspects

Termination Conditions

When do we stop? We need a computable, meaningful criterion.

Common criterion:

gradient mapping residual

$$\left\| G_\alpha(x^k) \right\| = \frac{1}{\alpha} \left\| x^k - x^{k+1} \right\| \leq \varepsilon$$

Additional/complementary conditions:

- Iterate change: $\left\| x^k - x^{k-1} \right\| \leq \varepsilon \max(1, \left\| x^k \right\|)$
- Objective change: $|\varphi(x^k) - \varphi(x^{k-1})| \leq \varepsilon \max(1, |\varphi(x^k)|)$
- Dual residual (g convex): $\left\| x^k - \mathrm{prox}_g(x^k - \nabla f(x^k)) \right\| \leq \varepsilon$
- Max iterations: $k \geq k_{\max}$

Practice: combine stationarity measure with iterate/objective change for robustness.

Termination Conditions (to be precise)

When do we stop? We need a computable **measure of optimality/stationarity**.

From the PGM update we have

$$0 \in \partial g(x_{k+1}) + \frac{x_{k+1} - x_k}{\alpha_k} + \nabla f(x_k)$$

$$\iff \quad \nabla f(x_{k+1}) - \nabla f(x_k) - \frac{x_{k+1} - x_k}{\alpha_k} \in \partial g(x_{k+1}) + \nabla f(x_{k+1}) = \partial \varphi(x_{k+1})$$

Stationarity criterion:

$$\operatorname{dist}(0, \partial \varphi(x_{k+1})) \overset{\text{PGM}}{\leq} \left\| \nabla f(x_{k+1}) - \nabla f(x_k) - \frac{x_{k+1} - x_k}{\alpha_k} \right\| \leq \varepsilon$$

Relaxing the Global Lipschitz Assumption

Global Lipschitz is often too strong:

- Unknown L, or L is very large (loose bound)
- Barrier / Nonconvex f: gradient may not be globally Lipschitz

Relaxations:

1. **Local Lipschitz + backtracking:** backtracking finds local L_k, guaranteed to terminate.
2. **Local descent lemma:** only require the backtracking condition (no global L).
3. **Proximal smoothness:** f prox-regular; weaker than convexity.
4. **Relative smoothness** [BBT17]: replace $\frac{1}{2}\|y-x\|^2$ by $D_h(y,x)$ (Bregman divergence); leads to *mirror descent / Bregman proximal gradient*.

Backtracking regularization

PGM with backtracking:

- prox-grad step: compute $x_+ \in \mathrm{prox}_{\alpha g}(x - \alpha \nabla f(x))$
- sufficient decrease: check if $\varphi(x_+) \leq \varphi(x_k) - \frac{\sigma}{\alpha} \|x_+ - x\|^2$
- backtracking: update α or break

For **nonconvex** f, backtracking is the standard practical approach. It is an adaptive (quadratic) regularization, not a linesearch procedure.

Even for **convex** f and g, adaptive schemes without checking descent [MM20, MM24, LTSP25]

Nonmonotone Line Search

Observation: Forcing $\varphi(x_{k+1}) \leq \varphi(x_k)$ every step can be overly restrictive

Key idea: replace *monotone* sufficient decrease condition with

$$\varphi(x_{k+1}) \leq \Phi_k - \frac{\sigma}{\alpha_k} \|x_{k+1} - x_k\|^2$$

where $\Phi_k \geq \varphi(x_k)$ and $\{\Phi_k\}$ ensures convergence.

Max-type — Grippo, Lampariello & Lucidi [GLL86]

Given $M \in \mathbb{N}$ and $m(k) := \min\{M, k\}$,

$$\Phi_k := \max_{0 \leq j \leq m(k)} \varphi(x_{k-j})$$

Mean-type — Zhang & Hager [ZH04], [DM23]

Given $\eta \in (0, 1]$,

$$\Phi_k := (1-\eta)\Phi_{k-1} + \eta\varphi(x_k), \quad \Phi_0 := \varphi(x_0)$$

Benefits:

- Accepts larger steps; avoids zigzagging
- Bounded nonmonotonicity; specializes to monotone
- Convergence guaranteed, possibly under additional assumptions (max...) [KM22, DM23]

Spectral (Barzilai–Borwein) Stepsizes

Idea: estimate the local Lipschitz constant from secant information.

Let $s_k := x_k - x_{k-1}$, $y_k := \nabla f(x_k) - \nabla f(x_{k-1})$. Mimic a diagonal quasi-Newton step:

BB Stepsizes — Barzilai & Borwein [BB88]

$$\alpha_k^{\mathrm{BB1}} = \frac{\langle s^k, s^k \rangle}{\langle s^k, y^k \rangle} \quad \text{(short step)}, \qquad \alpha_k^{\mathrm{BB2}} = \frac{\langle s^k, y^k \rangle}{\langle y^k, y^k \rangle} \quad \text{(long step)}$$

- Approximate the inverse of the average curvature
- In practice, must pair with **nonmonotone** globalization
- Dramatic **speedup** in practice (close to quasi-Newton)

Spectral Proximal Gradient Method

Algorithm: SPG [BMR00, JKMW23] / SpaRSA [WNF09]

x_0, $\alpha_0^{\mathrm{BB}} > 0$, $0 < \alpha_{\min} < \alpha_{\max}$
for $k = 0, 1, 2, \ldots$ **do**
    Starting from α_k^{BB}, by backtracking find α_k that delivers suff. decrease
    $x_{k+1} \in \operatorname{prox}_{\alpha_k g}(x_k - \alpha_k \nabla f(x_k))$
    Compute $s_{k+1} := x_{k+1} - x_k$, $y_{k+1} := \nabla f(x_{k+1}) - \nabla f(x_k)$
    $\alpha_{k+1}^{\mathrm{BB}} \leftarrow \operatorname{clip}(\alpha^{\mathrm{BB}}(s_{k+1}, y_{k+1}), \alpha_{\min}, \alpha_{\max})$
end for

In practice, often faster than FISTA

Toward (quasi-)Newton methods

Limitations of First-Order Proximal Gradient

First-order methods: only use ∇f (gradient information).

Issues:

- Rate $\mathcal{O}(1/k)$ (or $\mathcal{O}(1/k^2)$ with momentum) — slow for ill-conditioned problems
- Strong convexity gives linear rate $\mathcal{O}((1-\mu/L)^k)$ but $\mu/L \ll 1$ in practice

Newton-type methods: use curvature (Hessian or approximation).

- Replace $\frac{1}{2\alpha}\|x-x_k\|^2$ by $\frac{1}{2}(x-x_k)^\top H_k(x-x_k)$ in the model of f
- *Variable metric prox-gradient:*

$$x_{k+1} \in \operatorname{prox}_g^{H_k}(x_k - H_k^{-1}\nabla f(x_k))$$

- Challenge: $\operatorname{prox}^{H_k}$ may not be easy to compute for arbitrary H_k

Proximal Newton-type methods for convex structured minimization [PB13, LSS14]

Variable Metric Proximal Operator

Definition.

For SPD matrix H, the H-proximal operator is:

$$\operatorname{prox}_g^H(z) := \arg\min_u \left\{ g(u) + \tfrac{1}{2}(u-z)^\top H(u-z) \right\}$$

Key issue: for general H, the above may have no closed form even when prox_g does.

Exceptions (tractable prox_g^H):

- $g = \lambda \|\cdot\|_1$ and *diagonal* $H \implies$ scaled soft-threshold
- $g = \delta_C$ and diagonal $H \implies$ scaled projection
- g separable and H diagonal $\implies$ coordinate-wise prox

Full (quasi-)Newton requires more sophisticated approaches to be practical.

PANOC — Stella et al. [STSP17, TSP18]

Idea: combine a fast direction (quasi-Newton) with a safe PGM direction.
But, what about globalization?

Key observation.

The Moreau envelope is continuous.

PANOC algorithm:

1. Compute $x_{k+\frac{1}{2}} = \mathrm{prox}_{\alpha g}(x_k - \alpha \nabla f(x_k))$ (PGM step as anchor)
2. Build quasi-Newton direction on $\varphi_\alpha^{\mathrm{FB}}$ using L-BFGS
3. Linesearch to find x_{k+1} satisfying $\varphi_\alpha^{\mathrm{FB}}(x_{k+1}) \leq \varphi_\alpha^{\mathrm{FB}}(x_k) - \frac{\sigma}{\alpha} \|x_{k+1} - x_k\|^2$

Forward-Backward Envelope (FBE)

Definition [PB13].

The *forward-backward envelope* of $\varphi := f + g$ is:

$$\varphi_\alpha^{\mathrm{FB}}(x) := \min_u \left\{ f(x) + \langle \nabla f(x), u - x \rangle + \tfrac{1}{2\alpha} \|u - x\|^2 + g(u) \right\}$$

Properties (for suff. small $\alpha > 0$):

- $\varphi_\alpha^{\mathrm{FB}}(x) \leq \varphi(x)$ for all x
- $\varphi_\alpha^{\mathrm{FB}}(x) = \varphi(x) \iff x$ is a fixed point / critical / stationary
- for convex g, gradient mapping $G_\alpha(x) = \alpha \nabla \varphi_\alpha^{\mathrm{FB}}(x)$

PANOC minimizes $\varphi_\alpha^{\mathrm{FB}}$ by combining fast smooth quasi-Newton extrapolation steps with safe prox-grad updates.

PANOC: Properties and Convergence [STSP17, TSP18]

Structure:

- One prox evaluation per iteration
- One gradient evaluation per iteration
- L-BFGS memory update (low cost)
- Falls back to PGM if quasi-Newton step rejected

Convergence:

- $\left\|G_\alpha(x^k)\right\| \to 0$ (nonconvex)
- Superlinear rate near non-degenerate solutions
- Formally: R-superlinear under second-order sufficient conditions

PANOC$^+$: extension to locally Lipschitz ∇f, also with inexact prox evaluations [DMT22]

Applications: embedded optimal control, MPC, robotics. Subsolver in `OpEn` [SFP20], `alpaqa` [PSP22], `Bazinga` [DMJKM23].

Regularized Proximal Newton-type Method

Since [PB13, LSS14], form the **second-order structured model** at iterate x_k:

$$m_k(x) := f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2}(x - x_k)^\top B_k (x - x_k) + g(x)$$

where $B_k \approx \nabla^2 f(x_k)$ (exact Hessian or approximation) with regularization.

Proximal Newton-type subproblem

Solve

$$x_{k+1} \in \arg\min_x m_k(x)$$

Newton-type subproblem hardly available in closed-form, but a structured minimization: call PGM to solve it!

Ideas for practical proximal Newton-type methods: RPQN

RPQN [BFO19, Lec22]

Limited-memory quasi-Newton B_k with compact representation
$\implies$ small (memory M) system of equations to evaluate the scaled prox operator.

Nocedal's "Updating quasi-Newton matrices with limited storage" (1980)
Becker, Fadili & Ochs's "On quasi-Newton forward-backward splitting: Proximal calculus and convergence" (2019)
Kanzow & Lechner's "Efficient regularized proximal quasi-Newton methods for large-scale nonconvex composite optimization problems" (2024)

Ideas for practical proximal Newton-type methods: R2N

R2N [ABO22, DHO26]

- Solve the Newton-type proximal subproblem inexactly,
- ... obtaining at least the improvement of a prox-grad step (Cauchy decrease).
- Check if x_{k+1} delivers sufficient decrease for φ.
- If needed, recompute with an increased quadratic regularization.

Diouane, Habiboullah & Orban's "A proximal modified quasi-Newton method for nonsmooth regularized optimization" (2026)

Comparing the Methods: A Unified View

All methods minimize a *model* $m_k(x) \approx f(x) + g(x)$:

Method	**Model** $m_k(x)$	**Per-iter cost**
PGM	$\nabla f^\top d + \frac{1}{2\alpha}\|d\|^2 + g$	1 grad, 1 prox
PGM + BB	same, $\alpha_k = \alpha^{\mathrm{BB}}$, nonmono	1 grad, 1 prox
FISTA	same + momentum	1 grad, 1 prox
PANOC	envelope $\varphi_\alpha^{\mathrm{FB}}$ + L-BFGS	1 grad, 1 prox, QN update
R2N	$\nabla f^\top d + \frac{1}{2} d^\top B_k d + g$	inexact inner PGM solve
RPQN	same with compact represent.	1 grad, 1 inner root-finder

Trade-off: richer model $\implies$ fewer outer iterations but more expensive subproblem.
The *right* method depends on the problem structure, cost of g, Hessian availability.

In practice, choose solver depending on most/least expensive oracles at hand.
Some comparisons in [DHO26].

Numerical examples

Sparse linear regression with ProximalAlgorithms.jl / RegularizedOptimization.jl

$$\underset{x}{\text{minimize}} \quad \underbrace{\frac{1}{2}\|Ax - b\|^2}_{=:f(x)} + \lambda R(x)$$

$$\nabla f(x) = A^\top (Ax - b)$$

$$\nabla f(x) - \nabla f(y) = A^\top A(x - y)$$

[link to tutorial: PG]

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